



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

DIFFERENTIAL AND INTEGRAL CALCULUS 1M — 104018 WINTER 2021

Midterm

November 20, 2021

Student ID Number:

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Instructions

1. **Duration: 180 minutes.**
2. Please type your name and ID number at the top of each page. [UNNECESSARY, FIRST PAGE IS ENOUGH, MOST OF THEM FORGET AFTER THE SECOND PAGE ANYWAY]
3. This exam has ?? questions for a total of 100 points.
4. Write your answers clearly, completely and concisely in the space provided. Quote any results that you use. **No credit will be given to answers which are not justified.**
5. Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed. You will be provided with a dictionary (by demand) on the examination class (by one of the proctors).

ID number: _____

Question 1 (20 points) Let $f, g, h : I \longrightarrow \mathbb{R}$ be functions defined on an open interval I and $a \in I$. Suppose that:

- (i) $f(x) \leq g(x) \leq h(x)$, for all $x \in I$;
- (ii) $f(a) = h(a)$;
- (iii) $f'(a) = h'(a)$.

Prove that g is differentiable at a and that $g'(a) = f'(a) = h'(a)$.

Question 2 (30 points) Let $f : \mathbb{R} \longrightarrow \mathbb{R}$.
$$x \longmapsto e^{x^3} - 1$$

- (a) Show that f is one-to-one.
- (b) Compute $\text{im}(f) = \{f(a) \mid a \in \mathbb{R}\}$, the image (or range) of f .
- (c) Determine $f^{-1}(x)$, for all $x \in \text{im}(f)$, where $f^{-1} : \text{im}(f) \longrightarrow \mathbb{R}$ is the inverse of f .
- (d) Compute the tangent line to the graph of f at the point $(0, 0)$.

ID number:

Question 3 (20 points)

1. Let f be an everywhere continuous function. Show that if $\lim_{x \rightarrow a} f(x) = L < 0$ then there exists a neighborhood of $x = a$ such that $f(x) < 0$ for any x in that neighborhood. (10 p.)
2. Show, by providing an example, that the previous statement fails if f is not everywhere continuous. (10 p.)

ID number:

Question 4 (30 points)

Let f be a continuous function defined on the interval $(a, b) \subset \mathbb{R}$.

1. Show that if f is monotonic strictly increasing on (a, b) then f is 1-1 on (a, b) . (10 p.)

2. ‘ Assume that f is differentiable on (a, b) . Prove:

$$f'(x) > 0 \forall x \in (a, b) \Rightarrow f \text{ is monotonic strictly increasing on } (a, b).$$

(10 p.)

3. **True or False**

If f is continuous and monotonic strictly increasing on (a, b) , then f is differentiable on (a, b) . (10 p.)

(If you think that the statement is true then prove it, if you think the statement is false then provide a counter example).